Drone Kit



Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Micro Quadcopter kit; Plastic Hexacopter with Propulsion Kit; Quadcopter drone Combo with Pix hawk Kit for beginner

What it is used for

A drone kit provides components to assemble, configure, and operate a small unmanned aircraft for learning or approved tasks. Correct assembly, battery handling, calibration, and responsible flight practices help prevent crashes and injuries.

Main parts

- Frame supports the flight controller, motors, and other components.
- Brushless motors drive the propellers.
- Propellers generate lift and must match the motor specifications.
- Electronic speed controllers regulate motor speed on applicable kits.
- Flight controller processes sensor data and stabilises the aircraft.
- Battery supplies electrical power to the drone.
- Remote controller and receiver provide flight commands and communication.

Before you start

1. Read the kit manual and confirm all components are compatible.
2. Assemble the drone only under suitable supervision and with power disconnected.
3. Inspect the frame, propellers, motors, wiring, connectors, and battery for damage.
4. Check that propellers are fitted in the correct positions and orientations.
5. Charge the battery with a compatible charger on a safe, supervised surface.
6. Check local drone rules, permitted airspace, weather, and the approved flight area.

How to use

1. Secure all frame parts, motors, flight controller, and electronic connections according to the assembly guide.
2. Check motor and propeller compatibility and tighten propeller fittings as instructed.
3. Connect the flight controller and receiver using the specified wiring diagram.
4. Power on the controller and drone in the sequence required by the manufacturer.
5. Complete required calibration and configure failsafe settings according to the manual.
6. Check battery level, control response, sensor status, and communication before flight.
7. Place the drone on a level, open surface and keep people clear of the take-off area.
8. Start the motors and lift off gradually while maintaining visual line of sight.
9. Fly at a controlled height and speed within the permitted area and monitor battery status.
10. Land promptly when battery warnings appear or control becomes unreliable, then disarm the motors.

⚠ Safety

- Never install or remove propellers while the battery is connected.
- Keep hands, hair, and clothing away from rotating propellers.
- Do not fly near people, roads, power lines, airports, or restricted areas without legal authorisation.
- Never fly beyond visual line of sight unless specifically permitted by applicable rules.
- Do not use swollen, leaking, punctured, or overheating batteries.
- Stop flight during unsafe wind, rain, poor visibility, or loss of control.

Care & storage

- Disconnect the battery before cleaning, repairing, or changing components.
- Inspect propellers, motors, frame, wiring, and battery after each flight.
- Store batteries according to the manufacturer's charge and temperature guidance.
- Keep the drone and kit in a dry protective case and replace damaged parts before flying again.

Common mistakes to avoid

- Connecting the battery before checking wiring and propeller installation.
- Flying without checking calibration, battery level, and control response.
- Operating outside permitted airspace or losing visual contact with the drone.
- Continuing flight after battery warnings, damage, or unstable behaviour.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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